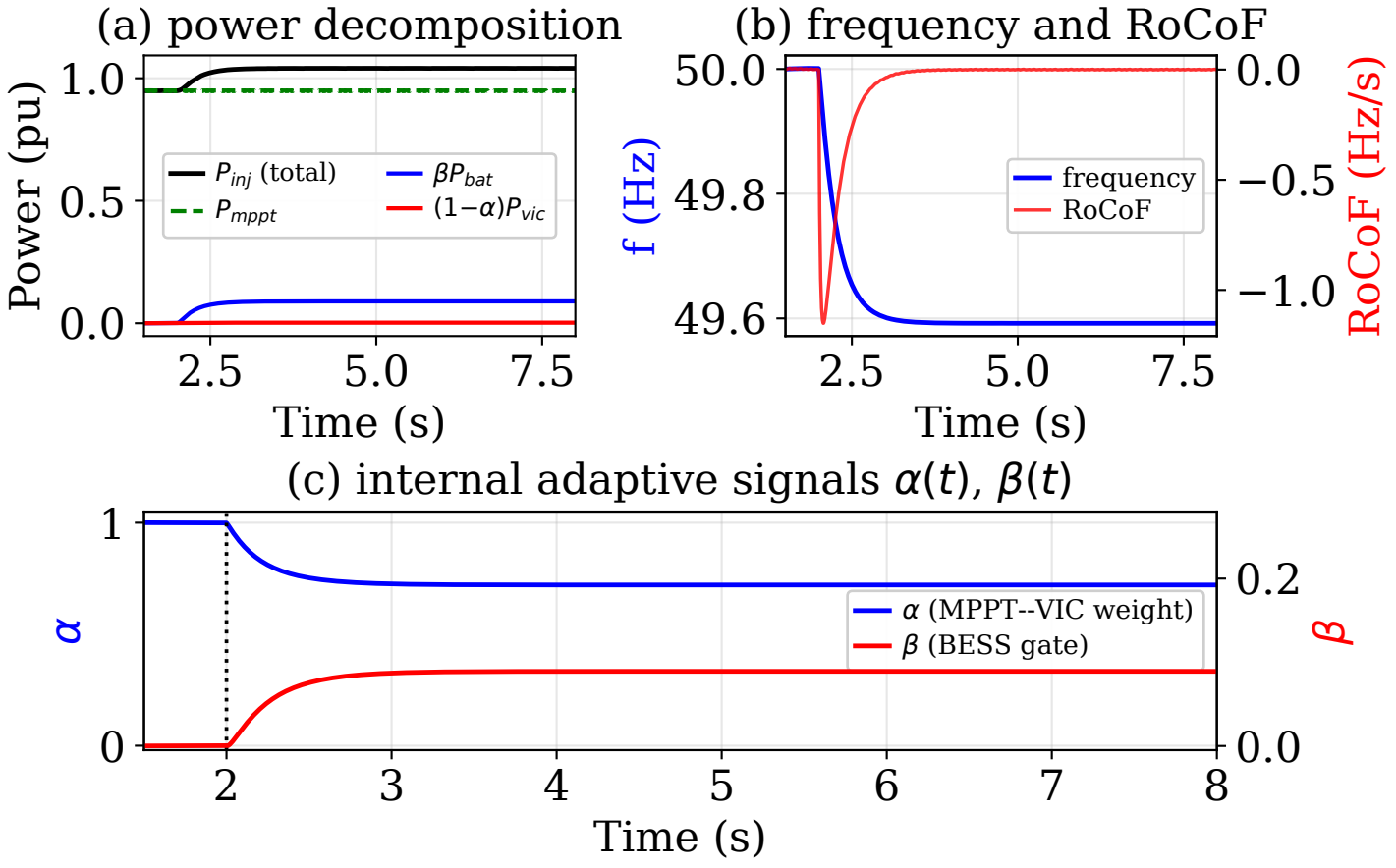
(a) power decomposition
Power (pu)
1.0
0.5
0.0
P_{inj} (total)
P_{mppt}
βP_{bat}
$(1-\alpha)P_{vic}$
2.5
5.0
7.5
Time (s)
(b) frequency and RoCoF
f (Hz)
50.0
49.8
49.6
frequency
RoCoF
0.0
−0.5
−1.0
RoCoF (Hz/s)
2.5
5.0
7.5
Time (s)
(c) internal adaptive signals $\alpha(t)$, $\beta(t)$
α
1
0
α (MPPT--VIC weight)
β (BESS gate)
0.2
0.0
β
2
3
4
5
6
7
8
Time (s)